Unsupervised identification of events in Chicago air-quality data

Feature Importance and K-Means Clustering of Fine Particulates, Gases, and Weather

Brandon Swanson, SULI Intern, Environmental Sciences Division
Mentors: Bhupendra Raut and Joseph O'Brien

Argonne National Laboratory

ABSTRACT

We applied K-means clustering to an averaged 30-minute rooftop measurements of particulate matter (PM) like $PM_{1.0}$, $PM_{2.5}$, PM_{10}, gases like CO, NO, NO_2, O_3, and meteorological components like temperature, dew point, U/V wind collected at Northeastern Illinois University (NEIU) from May through August 2023 to objectively parse air-quality regimes. The algorithm isolated five distinct states – Smog, Daytime, Background, with uniformly low pollutants, Smoke, with wildfire influence with elevated PM and CO, and Nocturnal – capturing both anthropogenic and smoke plume signals. Visualization tools such as standardized treemaps, diurnal frequency plots, and annotated time series clarified which variables drove separation and when each regime dominated, enabling discrimination between local emissions, regional background, and meteorological effects. Notably, the approach distinguished the sustained multi-day haze from Canadian wildfires – marked by persistently high PM and CO – from the brief particulate spike tied to July 4th fireworks. Diurnal composites further revealed midday ozone maxima under strong insolation and morning/evening NO/CO surges from traffic atop a steady urban baseline.

1. INTRODUCTION

Air quality monitoring typically focuses on particulate matter (PM) like $PM_{1.0}$, $PM_{2.5}$, PM_{10}, and trace gases such as CO, NO, NO_2, and O_3 because their near-surface concentrations directly influence human exposure and outdoor safety. During summer 2023, multiple Canadian wildfire plumes advected into Northeastern Illinois, producing some of the region's worst observed air quality and underscoring the need to objectively parse "smoke days" from more routine urban pollution. Beyond wildfire impacts, short-lived spikes tied to local activities – such as the sharp particulate surge around 4 July fireworks – can obscure broader patterns if not explicitly distinguished. Consequently, systematically classifying periods based on multivariate air-quality measurements is essential for interpretation, communication, and subsequent policy or public-health applications.

Prior research has documented diurnal pollutant cycles driven by photochemistry and traffic, yet urban datasets often blend regional smoke intrusions, anthropogenic emissions, and meteorological modulation without a transparent, data-driven partition. Daily ozone maxima under strong insolation can coincide with morning and evening NO/CO surges from commuting patterns, all superimposed on a relatively steady "background" signal. What remains unresolved is how to disentangle these overlapping signals – and transient anomalies like fireworks – from each other in a reproducible way, and how meteorological structure conditions those regimes.

This study produced an open-source, high-frequency rooftop dataset[1] collected at Northeastern Illinois University (NEIU) from May through August 2023, encompassing fine particulates, gases, and coincident meteorological variables. Measurements were acquired using Vaisala AQT air-quality and WXT weather sensors, providing temperature, dew point, humidity, rain rate, wind speed, and direction alongside pollutant concentrations. Data were rigorously preprocessed – invalid values masked, records averaged to 30 minutes, and features standardized to zero mean and unit variance – to ensure comparability across variables prior to analysis.

2. METHODS

Continuous air-quality and meteorological measurements were obtained from the NEIU CROCUS rooftop site in Chicago, Illinois, for the period of May through August 2023. Two co-located Vaisala instruments provided the raw observations: an AQT module for particulate matter ($PM_{1.0}$, $PM_{2.5}$, PM_{10}) and trace gases (CO, NO, NO_2, O_3), and a WXT sensor for temperature, dew point, relative humidity, rain rate, wind speed, and wind direction. Instruments sampled in high-frequency intervals, yielding a dataset suitable for resolving diurnal variability and short-lived pollution episodes like wildfire smoke intrusions and holiday fireworks.

Quality control and preprocessing followed a four-step sequence. First, physically implausible or flagged values, like negative concentrations, sensor error codes, or unusual data, were masked. Second, the wind speed and direction were used to produce U and V winds. This was to maintain likeness of the wind to other variables without large numbers, like wind direction. Third, to reduce noise and align all variables on a common timescale, data were averaged to 30-minute means. Fourth, each variable was standardized to zero mean and unit variance to prevent high-magnitude features, such as PM spikes, from dominating distance calculations in the clustering step. Records with more than a minimal fraction of missing variables after aggregation were discarded rather than imputed to avoid introducing artificial structure.

Unsupervised classification was performed with K-means clustering. After standardization, Euclidean distance to candidate centroids was minimized using the scikit-learn implementation (random_state fixed for reproducibility, n_init ≥ 20, max_iter = 300). The number of clusters, k, was selected through an iterative process: (1) computing within-cluster sum of squares across k = 1–14 and inspecting the "elbow" in that curve; (2) examining silhouette coefficients; and (3) visually assessing whether candidate solutions produced meteorologically interpretable regimes. The final solution of k = 5, balanced statistical compactness with interpretability, yielding clusters later labeled Smog, Daytime, Background, Smoke, and Nocturnal based on their characteristic pollutant–meteorology signatures.

Post-clustering diagnostics and visualization were essential for verifying cluster quality and assigning descriptive names. Distributional plots, such as histograms and boxplots, highlighted how each variable varied across clusters; pairplots exposed inter-variable relationships; and treemaps of cluster centroids [Figure 1] summarized relative variable magnitudes in a compact, standardized format.

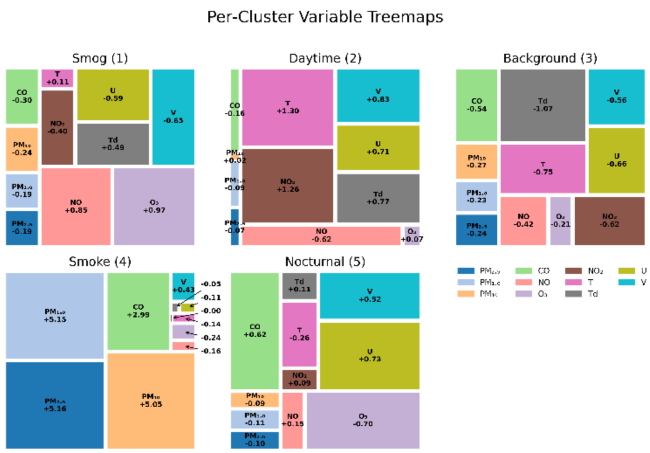

Figure 1: Variable treemaps for each of the five clusters, where the tile area reflects the standardized deviation of each pollutant or meteorological variable from its overall mean. The negative values are below the standard deviation and positive values are above the standard deviation

Additional tools included diurnal frequency plots – cluster occurrence by local time – and annotated time series to confirm that the "Smoke" cluster aligned with known Canadian wildfire plume arrivals and that short-lived particulate matter surges around 4 July were captured distinctly.

All processing and analyses were conducted in Python and Jupyter Notebooks utilizing pandas, numpy, matplotlib, scikit-learn, and more within a reproducible notebook workflow. Scripts are organized to allow another researcher to (1) ingest sensor netCDF files, (2) apply identical filters and scaling, (3) rerun K-means with chosen parameters, and (4) regenerate all figures. This level of detail is intended to ensure full replicability and facilitate extension to other urban air-quality datasets or additional clustering algorithms.

3. RESULTS

A composite visualization [Figure 2] integrating a bar chart of relative frequencies, an hourly occurrence curve, and a full-season timeline revealed five clearly differentiated regimes – smog, daytime, smoke, background, and nighttime – each with distinct temporal behavior. Light-gray shading on the timeline denoted wildfire smoke periods, red bands marked the most intense smoke episodes, and a dashed line highlighted the 4 July fireworks spike. Initial inspection of pollutant patterns streamlined cluster assignment: plotting variable distributions and temporal signatures made the separation of regimes more evident before any labels were applied.

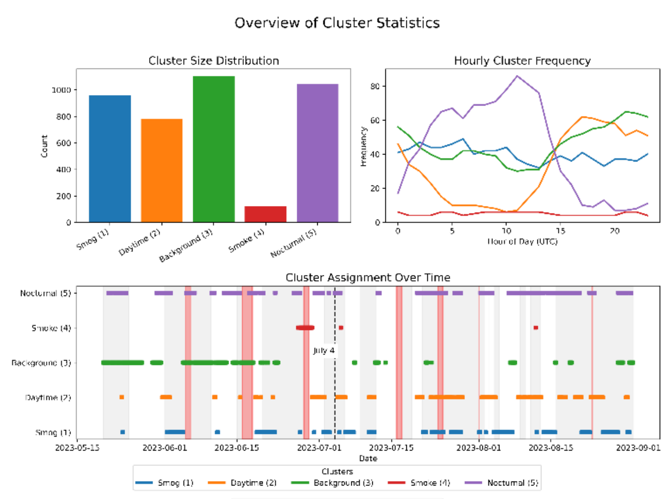

Figure 2: This composite view summarizes the five clusters: the bar chart (top left) shows each cluster's relative frequency; the line plot (top right) displays their average hourly occurrence (UTC), highlighting smog, daytime, smoke, background, and nighttime signatures; and the timeline (bottom) displays cluster

assignments over May 20, 2023 through August 30, 2023, with light gray shading marking wildfire smoke periods, red bands indicating intense smoke events, and a dashed line on July 4 for fireworks

Cluster 4, “Smoke”, stood out through persistently elevated particulate matter and CO, consistent with the multi-day Canadian wildfire haze that affected Chicago. In contrast, the 4 July particulate surge appeared as a short, sharp anomaly, confirming that the clustering distinguished episodic fireworks impacts from longer smoke events. Diurnal structure was quantified using the frequency – time plot: clusters exhibited preferred hours, with some regimes peaking overnight and others during the afternoon photochemical maximum. This visualization was pivotal for confirming that temporal occurrence, not just concentration magnitude, contributed to regime identity.

Per-cluster treemaps and boxplots [Figures 1 and 3] resolved ambiguous cases by summarizing standardized deviations and full distributions, respectively. Treemaps quickly conveyed which variables dominated each centroid, while boxplots exposed spread and outliers across PM, gases, and meteorological fields, thereby validating the internal consistency of each cluster.

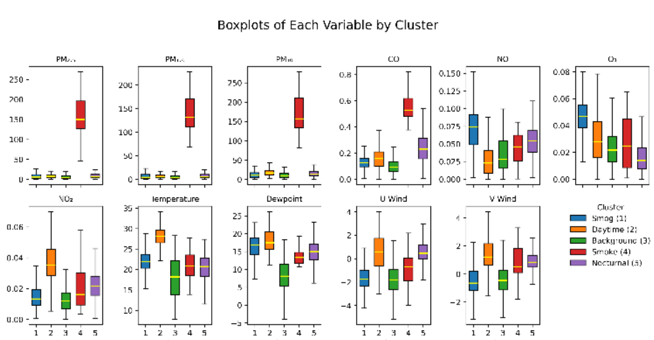

Figure 3: Boxplots of air quality and meteorological variables by cluster, illustrating distributions in particulate matter, gases, temperature, dewpoint, and wind components. Each cluster’s unique signature – ranging from intense smoke and smog to nocturnal and background conditions – is clearly differentiated by the position and spread of the boxes

Cluster 1, “Smog”, featured comparatively high NO and O_3, a signature of strong photochemical processing amid urban emissions. This pairing distinguished it from smoke events – high PM/CO – and from nocturnal or background regimes where ozone was suppressed.

Cluster 3, “Background”, maintained uniformly low pollutant values across variables, serving as the baseline against which other regimes. Its narrow interquartile ranges in the boxplots reinforced its stability and low variability.

Broader pollutant cycles aligned with established urban chemistry: midday ozone peaks under strong sunlight coincided with morning/evening NO and CO surges from traffic, all superimposed on the background regime.

Overall, the clustering separated sustained wildfire smoke, routine photochemical smog, diurnally driven daytime and nocturnal conditions, and a clean background. Each regime’s uniqueness was evident in both central tendency (centroids/medians) and variability (box spread), confirming that the five-cluster solution balanced interpretability with statistical compactness.

4. DISCUSSION

The K-means clustering framework effectively separated multiple, overlapping air-quality regimes in Northeastern Illinois during summer 2023, yielding clear insights into both episodic and systematic pollutant patterns. By isolating a “Smoke” cluster (Cluster 4) marked by persistently elevated PM and CO over several days, the analysis captured the sustained impact of Canadian wildfire plumes on regional air quality. In contrast, the brief but intense 4 July fireworks spike appeared as a distinct, high-PM anomaly that, without clustering, might have been mixed with more prolonged smoke events. This separation is critical for public-health advisories: sustained haze warrants different messaging – like multi-day activity restrictions – than a short, localized pollution spike from fireworks.

Beyond extreme events, the five-cluster solution illuminated the underlying diurnal cycle of urban pollutant dynamics. “Daytime” and “Smog” clusters revealed the progression of morning NO/CO emissions from traffic, a midday ozone peak driven by photochemistry, and evening rush-hour spikes. The “Nocturnal” cluster – characterized by elevated CO but suppressed O_3 – highlighted boundary-layer collapse and nocturnal inversion formation, during which pollutants accumulate near the surface. The “Background” regime, with uniformly low concentrations and narrow variability, provided a baseline against which excursions could be measured. Together, these patterns corroborate established urban-chemistry patterns (traffic emissions + photochemical ozone production) while quantifying their seasonal behavior in a single, high-frequency dataset.

The implications of these findings are diverse. First, clustering offers an automated, data-driven tool for real-time air-quality classification, enabling environmental agencies to tailor advisories dynamically – distinguishing days with wildfire influence from routine smog episodes or atypical pollution sources like fireworks. Second, the approach provides a foundation for integrating air-quality regimes into epidemiological studies, by linking health outcomes (e.g., asthma attacks) to specific pollutant signatures rather than aggregate indices. Finally, the methodology is extensible to other urban environments and sensor networks, supporting comparative studies of emission profiles and regional transport phenomena.

Nonetheless, several limitations warrant consideration. K-means assumes clusters of roughly equal, spherical shape in standardized space; regimes with more complex or non-convex boundaries might be better captured by density-based methods or Gaussian mixture models. Additionally, the rooftop location provides a single point of measurement, as other locations were unusable for this study; spatial assortment across the urban canopy layer remains unexplored.

Future research should address these limitations by (1) comparing clustering results with alternative algorithms and temporal resolutions; (2) integrating vertical profiling data (e.g., lidar or balloon soundings) to relate regimes to boundary-layer height and stability; and (3) extending the temporal record across multiple seasons and years to assess interannual variability and long-term trends. Coupling unsupervised classification with chemical-transport modeling would further enhance understanding of source contributions and regional pollutant transport. By refining these approaches, researchers and policymakers can develop more robust, targeted strategies for air-quality management and public-health protection.

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[1] tinyurl.com/Chicago-Air-Quality-SULI